HR ANALYTICS

Employee Attrition & Retention Analysis

Business Analysis Report | IBM HR Analytics Employee Attrition & Performance dataset

Project objective

This project examines employee attrition patterns in a simulated HR dataset. It uses data preparation, exploratory analysis, SQL queries and an interactive Power BI dashboard to identify employee groups and workplace factors associated with attrition, and to develop practical questions and recommendations for HR decision-makers.

1. Executive Summary

The analysis covers 1,470 employee records and reports an overall attrition rate of **16.12%** (237 employees). Attrition is not evenly distributed across the groups examined. Higher rates appear among employees with short company tenure, Sales Representatives, and employees recorded as working overtime. Employees who left also had a lower average monthly income than employees who stayed. These patterns can help HR prioritize further investigation, but they do not establish why employees left.

2. Business Problem and Objectives

Employee turnover can create recruitment, onboarding, training and productivity costs. Understanding how attrition differs across employee groups can help an organization decide where to investigate retention risks. The analysis therefore aims to:

- Measure overall attrition and retention.
- Compare attrition across departments, job roles and demographic groups.
- Examine associations with tenure, overtime, job satisfaction and monthly income.
- Present findings in a Power BI dashboard and propose evidence-informed follow-up actions.

3. Dataset Overview

The project uses the IBM HR Analytics Employee Attrition & Performance dataset, a simulated dataset commonly distributed for analytics practice. The original file contains 1,470 rows and 35 columns. Three constant columns—EmployeeCount, Over18 and StandardHours—were removed during preparation, leaving 1,470 rows and 32 columns in the cleaned dataset.

The dataset contains no missing values, duplicate rows or duplicate EmployeeNumber values in the checks performed. The results describe this dataset only and should not be treated as representative of a real organization or the wider labor market.

4. Methodology

Data preparation: inspected the data structure, checked missing values and duplicates, reviewed selected value ranges and removed constant columns.

Exploratory data analysis: used Python to summarize attrition by employee and workplace characteristics.
SQL analysis: loaded the cleaned data into SQLite and wrote queries using aggregation, grouping and conditional calculations to validate key metrics.
Visualization: built a Power BI dashboard with KPI cards, charts and slicers to explore attrition patterns.

5. Key Findings

Measure / group	Result
Total employees	1,470
Employees who left	237
Employees who stayed	1,233
Overall attrition rate	16.12%
Retention rate	83.88%
Average monthly income – stayed	6,832.74
Average monthly income – left	4,787.09

Tenure: The 0–1 year group had an attrition rate of 34.88%, compared with 8.13% among employees with 11+ years at the company. The pattern shows that attrition was highest among employees with the shortest tenure.

Job role: Sales Representatives had the highest role-level attrition rate at 39.76%. Laboratory Technicians were at 23.94%, while Research Directors had the lowest rate at 2.50%. Group sizes differ, so rates should be interpreted alongside employee counts.

Overtime: Employees recorded as working overtime had an attrition rate of 30.53%, compared with 10.44% for employees not recorded as working overtime.

Department: Sales had an attrition rate of 20.63%, Human Resources 19.05% and Research & Development 13.84%. The Human Resources group is relatively small (63 employees).

Job satisfaction: Attrition rates were 22.84% for satisfaction score 1, 16.43% for score 2, 16.52% for score 3 and 11.33% for score 4. The pattern suggests lower satisfaction scores were associated with higher attrition overall, although scores 2 and 3 were similar.

Monthly income: Average monthly income was 6,832.74 among employees who stayed and 4,787.09 among employees who left. This is a descriptive difference and may be related to other factors such as job role, seniority or experience.

6. Recommendations

1. Strengthen early-tenure support. Review onboarding, manager check-ins, role expectations and access to training during the first year. Track retention at defined tenure milestones.

2. Investigate overtime patterns. Review overtime frequency, duration, team distribution and workload planning. Discuss whether staffing or scheduling adjustments are needed.

3. Examine Sales Representative retention. Conduct role-specific follow-up analysis covering workload, targets, training, manager support, career progression and compensation structure.

4. Review compensation in context. Compare pay within similar roles, levels and experience bands rather than interpreting the overall income difference as evidence of a pay-related cause.

5. Monitor employee experience. Use regular, confidential feedback and satisfaction measures alongside attrition tracking to identify areas requiring investigation.

6. Establish a recurring HR dashboard review. Monitor attrition rate and employee counts by tenure, role, department and overtime status. Investigate changes over time before deciding on interventions.

7. Limitations

This is a simulated dataset, not verified organizational HR data. The analysis is descriptive and cross-sectional; it does not prove causation. Some categories contain relatively few employees, so their rates can be sensitive to small changes in counts. The dataset does not provide all potentially relevant context, and the analysis should not be used alone to make decisions about individual employees. Findings should be validated with current, organization-specific data and qualitative input before action is taken.

8. Conclusion

The analysis identified several groups with comparatively higher attrition rates, particularly employees in their first year, Sales Representatives and employees working overtime. It also found differences in average monthly income and attrition across job satisfaction levels and departments. The Power BI dashboard makes these patterns easier to explore, while the SQL and Python analyses provide supporting calculations. The results offer a starting point for targeted HR investigation rather than definitive explanations of employee turnover.

Suggested project tools: Python (Pandas, Matplotlib/Seaborn if used), SQLite, SQL, Power BI, Git and GitHub.

Dataset note: Cite the dataset's original Kaggle page in your GitHub README and check its license/usage terms before redistributing the CSV.